Module 6 slides: Repetitive IV Bolus and Intermittent IV Infusions (worked examples)

Module 6 - Multiple Dosings · Lecture slides, worked examples · 23 Sep 2026

Name _____ Date _____

Example 1 An antibiotic has an average t½ of approximately 4 hours and an apparent V_D that is 25% of body weight. The drug is to be administered 10 mg/kg every 8 hours by multiple IV bolus injections to a 65-kg female. What is the expected maximum concentration after the first dose?

Example 1 The same antibiotic (t½ 4 hours, V_D 25% of body weight, 10 mg/kg every 8 hours by IV bolus to a 65-kg female) peaks at 40 mg/L after the first dose. What is the expected minimum concentration after the first dose?

Example 1(a)

An antibiotic has an average t½ of approximately 4 hours and an apparent V_D that is 25% of body weight. The drug is to be administered 10 mg/kg every 8 hours by multiple IV bolus injections to a 65-kg female. What is the expected maximum concentration at steady state?

Example 1(b)

The same antibiotic (t½ 4 hours, V_D 25% of body weight, 10 mg/kg every 8 hours by IV bolus to a 65-kg female) reaches a maximum of 53.3 mg/L at steady state. What is the expected minimum concentration at steady state?

Example 1(c)

An antibiotic has an average t½ of approximately 4 hours and an apparent V_D that is 25% of body weight. The drug is to be administered 10 mg/kg every 8 hours by multiple IV bolus injections to a 65-kg female. What is the expected average concentration at steady state?

Example 2 An antibiotic has an average t½ of approximately 4 hours and an apparent V_D that is 25% of body weight. The drug is to be administered 10 mg/kg every 8 hours by multiple IV bolus injections to a 65-kg female. What is the expected concentration 3 hours after injection of the 2nd dose?

Example 3 An antibiotic has an average t½ of approximately 4 hours and an apparent V_D that is 25% of body weight. The drug is administered 10 mg/kg every 8 hours by multiple IV bolus injections to a 65-kg female, and the maximum at steady state is 53.3 mg/L. What is the expected concentration 3 hours after injection of the last dose (assuming steady state was attained)?

Slide "Plasma Drug Concentration at Steady State"

An antibiotic with a t½ of 4 hours is at steady state on repeated IV bolus doses, with a maximum of 53.3 mg/L. What is the expected concentration 4 hours after injection of the last dose?

Example 4 A 300-mg dose of an antibiotic is administered as an intravenous (IV) infusion over a period of 2 hours. What is the infusion rate?

Example 4(a)

A 300-mg dose of an antibiotic was administered as an intravenous (IV) infusion over a period of 2 hours ($k = 0.15\ hr^{-1}$, $V_D = 15$ L). What is the plasma drug concentration at the end of the first infusion?

Example 4(b)

A 300-mg dose of an antibiotic was administered as an intravenous (IV) infusion over a period of 2 hours. Six hours after the start of the first infusion, a second dose of 300 mg was infused, again over 2 hours ($k = 0.15\ hr^{-1}$, $V_D = 15$ L). Each infusion reaches 17.28 mg/L at its end. What is the plasma drug concentration 4 hours after the cessation of the second infusion?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

Example 1 Her key

Answer: 40 mg/L

WHAT THE STEM GIVES

$t_{1/2}$	approximately 4 hours	*not needed: the first-dose peak has no exponential*
V_D	25% of body weight	read as 0.25 L per kg, the denominator
D_0	10 mg/kg	the numerator, per kilogram
τ	every 8 hours	*not needed: no accumulation before the first dose*
weight	65-kg	*not needed: the kilograms cancel between dose and volume*

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_0 = \frac{D_0}{V_D}$$

For an intravenous (IV) bolus the whole dose is in the body at time zero, so the first-dose peak is the dose divided by the volume of distribution. No drug was present before this dose.

UNIT

$$C_0 = \frac{10\ \text{mg/kg}}{0.25\ \text{L/kg}}$$

Both the dose and the volume are given per kilogram, so the kilograms cancel and the patient's weight is not needed. 25% of body weight is 0.25 L per kg.

ALGEBRA

$$C_0 = 40\ \text{mg/L}$$

10 divided by 0.25 is 40, and mg over L leaves a concentration. The same result comes from 650 mg divided by 16.25 L.

ROUND

$$C_0 = 40\ \text{mg/L}$$

Her value, exact here. Every steady-state concentration in this example is this 40 mg/L multiplied by a factor that holds only k and τ, so an error here carries into every later part.

> **DOES THE NUMBER MAKE SENSE?**
> Ten milligrams spread through a quarter of a litre for every kilogram is 40 mg/L; the weight cancels. The first-dose peak is below the steady-state peak, which the accumulation factor raises.

Example 1 Her key

Answer: 10 mg/L

WHAT THE STEM GIVES

$t_{½}$	4 hours	gives $k = \frac{0.693}{4} = 0.1733\ hr^{-1}$; 8 hours is 2 half-lives
V_D	25% of body weight	*not needed: the peak is given*
D_0	10 mg/kg	*not needed: already inside the given peak*
τ	every 8 hours	the exponent kτ, with t = τ for the trough
weight	65-kg	*not needed*
C_0	40 mg/L	the first-dose peak the trough is read down from

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{min} = C_0 e^{-k\tau}$$

After the first dose the level falls by first-order elimination until the next dose, so the minimum is at the end of the interval, t = τ = 8 hr.

ALGEBRA

$$\frac{8\ hr}{4\ hr} = 2\ \text{half-lives}$$

The interval is exactly two half-lives, so the level halves twice: 40, then 20, then 10. Two half-lives leave one quarter of the peak, and 40 × 0.25 = 10 mg/L.

ALGEBRA

$$C_{min} = 40\ mg/L \times e^{-(0.693/4\ hr)(8\ hr)} = 40 \times 0.25 = 10\ mg/L$$

$k = \frac{0.693}{4}\ hr = 0.1733\ hr^{-1}$, and kτ = 1.386 with no units, so $e^{-1.386} = 0.25$. The hours cancel inside the exponent.

ROUND

$$C_{min} = 10\ mg/L$$

Her value, 10 mg/L. It is the first-dose trough, the level just before the second dose, and the steady-state trough will be higher than it by the same accumulation factor as the peak.

> **DOES THE NUMBER MAKE SENSE?**
> Eight hours is two half-lives of 4 hours, so the peak halves twice: 40 → 20 → 10 mg/L. The first-dose trough has to lie below the 40 mg/L peak.

Example 1(a) Her key

Answer: 53.3 mg/L

WHAT THE STEM GIVES

$t_{½}$	approximately 4 hours	gives $k = \frac{0.693}{4} = 0.1733\ hr^{-1}$
V_D	25% of body weight	with the dose gives $C_0 = \frac{10}{0.25} = 40\ mg/L$
D_0	10 mg/kg	the numerator of C_0
τ	every 8 hours	the exponent kτ = 0.1733 × 8 = 1.386
weight	65-kg	*not needed: the kilograms cancel*

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{max}^{\infty} = \frac{C_0}{1 - e^{-k\tau}}$$

At steady state each dose is added to drug left from earlier doses. Dividing the first-dose peak by $(1 - e^{-k\tau})$ adds that accumulated drug.

UNIT

$$C_0 = \frac{10 \text{ mg/kg}}{0.25 \text{ L/kg}} = 40 \text{ mg/L}; \; k = \frac{0.693}{4 \text{ hr}} = 0.1733 \text{ hr}^{-1}$$

The kilograms cancel in C_0. The rate constant comes from the half-life, so it is in reciprocal hours, matching τ in hours.

ALGEBRA

$$k\tau = (0.1733 \text{ hr}^{-1})(8 \text{ hr}) = 1.386; \; e^{-1.386} = 0.25$$

Reciprocal hours times hours leaves a pure number, which is what an exponent must be. 0.25 is the fraction of each dose left at the end of the interval.

ALGEBRA

$$C_{max}^{\infty} = \frac{40 \text{ mg/L}}{1 - e^{-(0.693/4)(8)}} = \frac{40 \text{ mg/L}}{1 - 0.25} = 53.3 \text{ mg/L}$$

Dividing by 0.75 raises the peak by a third, the accumulation factor 1.33 for an interval of two half-lives. 40 divided by 0.75 is 53.33.

ROUND

$$C_{max}^{\infty} = 53.3 \text{ mg/L}$$

Her value, 53.3 mg/L. It is greater than the first-dose peak of 40 mg/L, as a steady-state peak must be; a result below 40 means the factor was multiplied instead of divided.

> **DOES THE NUMBER MAKE SENSE?**
> Eight hours is two half-lives, so a quarter of each dose is still there when the next arrives. The steady-state peak is C_0 over $1 - 0.25$, a third above 40 mg/L: 53.3 mg/L, above 40 and below 2×40.

Example 1(b) Her key

Answer: 13.3 mg/L

WHAT THE STEM GIVES

t½	4 hours	gives $k = \frac{0.693}{4} = 0.1733 \text{ hr}^{-1}$; 8 hours is 2 half-lives
V_D	25% of body weight	*not needed: the steady-state peak is given*
D_0	10 mg/kg	*not needed: already inside the given peak*
τ	every 8 hours	the exponent $k\tau = 0.1733 \times 8$
weight	65-kg	*not needed*
C_{max}^{∞}	53.3 mg/L	the steady-state peak the trough is read down from

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{min}^{\infty} = C_{max}^{\infty} e^{-k\tau}$$

From the steady-state peak to the steady-state trough is one interval of first-order elimination, so $C = C_0 e^{-kt}$ with C_{max}^{∞} as the starting value and $t = \tau$. This is the same as the equation $\frac{C_0\, e^{-k\tau}}{1 - e^{-k\tau}}$.

ALGEBRA

$$C_{min}^{\infty} = 53.3 \text{ mg/L} \times e^{-(0.693/4 \text{ hr})(8 \text{ hr})} = 53.3 \times 0.25$$

k is unchanged at steady state, 0.1733 hr^{-1}, and $k\tau = 1.386$, a pure number. The interval is two half-lives, so the peak is halved twice.

ALGEBRA

$$C_{min}^{\infty} = 13.3 \text{ mg/L}$$

$53.3 \times 0.25 = 13.3$. Starting from the first-dose peak instead gives $\frac{40}{0.75} \times 0.25$, the same value by a longer route.

ROUND

$$C_{min}^{\infty} = 13.3 \text{ mg/L}$$

Her value. It is above the first-dose trough of 10 mg/L by the same factor, 1.33, as the peak is above 40 mg/L.

> **DOES THE NUMBER MAKE SENSE?**
> Eight hours is two half-lives of 4 hours, so a quarter of the 53.3 mg/L peak remains: $53.3 \times 0.25 = 13.3$ mg/L. A trough lies below its peak.

Example 1(c) Her key

Answer: 28.9 mg/L

WHAT THE STEM GIVES

$t_{½}$	approximately 4 hours	gives $k = \frac{0.693}{4} = 0.1733\ hr^{-1}$
V_D	25% of body weight	with the dose gives $\frac{D_0}{V_D} = 40$ mg/L
D_0	10 mg/kg	the numerator; F = 1 for IV
τ	every 8 hours	the denominator $k\tau = 1.386$
weight	65-kg	*not needed: the kilograms cancel*

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{avg}^{\infty} = \frac{FD_0}{V_D\, k\tau}, \text{ with } F = 1$$

The average over one interval at steady state is the dose divided by clearance times the interval, and clearance is V_D k. An intravenous (IV) dose has F = 1.

UNIT

$$\frac{D_0}{V_D} = \frac{10 \text{ mg/kg}}{0.25 \text{ L/kg}} = 40 \text{ mg/L}$$

$\frac{D_0}{V_D}$ is the first-dose C_0 again, so the kilograms cancel and the value from part one can be reused. The dose is 650 mg and V_D is 16.25 L, which give the same 40 mg/L.

ALGEBRA

$$C_{avg}^{\infty} = \frac{40 \text{ mg/L}}{(0.693/4 \text{ hr})(8 \text{ hr})} = \frac{40 \text{ mg/L}}{1.386}$$

k is $\frac{0.693}{4}$ = 0.1733 hr^{-1} and τ is 8 hr. k times τ is reciprocal hours times hours, a pure number, so the result stays in mg/L.

ALGEBRA

$$C_{avg}^{\infty} = 28.9 \text{ mg/L}$$

40 divided by 1.386 is 28.86. kτ is 1.386 because the interval is two half-lives: 2 × 0.693. No exponential is needed for the average.

ROUND

$$C_{avg}^{\infty} = 28.9 \text{ mg/L}$$

Her value, 28.9 mg/L. It lies between the steady-state trough, 13.3, and peak, 53.3, and below their midpoint, 33.3, because the decline is exponential.

> **DOES THE NUMBER MAKE SENSE?**
> C_{avg}^{∞} is C_0 = 40 mg/L divided by kτ = 1.386, so a little under three-quarters of C_0: 28.9 mg/L. It lies between the steady-state trough and peak, so above 10 mg/L and below 2 × 40 mg/L.

Example 2 Her key

Answer: 29.7 mg/L

WHAT THE STEM GIVES

t½	approximately 4 hours	gives k = $\frac{0.693}{4}$ = 0.1733 hr^{-1} for both exponents
V_D	25% of body weight	with the dose gives $\frac{D_0}{V_D}$ = 40 mg/L
D_0	10 mg/kg	the numerator of $\frac{D_0}{V_D}$
τ	every 8 hours	kτ = 1.386 in the accumulation factor
weight	65-kg	*not needed: the kilograms cancel*

n	2nd dose	n = 2 in $e^{-nk\tau}$
t	3 hours after injection	t = 3 hr in e^{-kt}

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_p = \frac{D_0}{V_D}\left(\frac{1 - e^{-nk\tau}}{1 - e^{-k\tau}}\right) e^{-kt}, \text{ with } n = 2 \text{ and } t = 3 \text{ hr}$$

Two doses are not steady state, so the n-dose equation is used. n is the dose just given and t is the time since it was given.

UNIT

$$\frac{D_0}{V_D} = 40 \text{ mg/L}; \; k = \frac{0.693}{4 \text{ hr}} = 0.1733 \text{ hr}^{-1}; \; k\tau = 1.386$$

The kilograms cancel in $\frac{D_0}{V_D}$, giving 40 mg/L. All times are in hours, so kτ = 0.1733 × 8 = 1.386 and each exponent is a pure number.

ALGEBRA

$$\frac{1 - e^{-2(1.386)}}{1 - e^{-1.386}} = \frac{1 - 0.0625}{1 - 0.25} = \frac{0.9375}{0.75} = 1.25$$

$e^{-2.772} = 0.25^2 = 0.0625$. The bracket is how far the peak has risen after two doses: 1.25 times C_0, against 1.33 at steady state.

ALGEBRA

$$e^{-kt} = e^{-(0.1733 \text{ hr}^{-1})(3 \text{ hr})} = e^{-0.520} = 0.5945$$

This is the decline for the 3 hours since the second dose. Three hours is less than one half-life, so the factor must lie between 1 and 0.5.

ALGEBRA

$$C_p = 40 \text{ mg/L} \times 1.25 \times 0.5945 = 29.7 \text{ mg/L}$$

40 × 1.25 = 50 mg/L is the peak just after the second dose: 40 from the new dose plus the 10 left from the first. Three hours of decline takes it to 29.7.

ROUND

$$C_p = 29.7 \text{ mg/L}$$

Her value, 29.7 mg/L. It is below the 31.7 mg/L found 3 hours after a dose at steady state, because after two doses less drug has accumulated.

> **DOES THE NUMBER MAKE SENSE?**
> Just after the 2nd dose the level is 40 plus the 10 mg/L left from the first, 50 mg/L; 3 hours is under one half-life, so the answer sits between 25 and 50 mg/L.

Example 3 Worked by the drill

Answer: 31.7 mg/L

WHAT THE STEM GIVES

t½	approximately 4 hours	gives $k = \frac{0.693}{4} = 0.1733 \text{ hr}^{-1}$

V_D	25% of body weight	*not needed: the steady-state peak is given*
D_0	10 mg/kg	*not needed: already inside the given peak*
τ	every 8 hours	*not needed: the time asked is 3 hours, within the interval*
weight	65-kg	*not needed*
C_{max}^{∞}	53.3 mg/L	the peak the level decays from
t	3 hours after injection	t in e^{-kt}

THE WORKING, ONE LINE AT A TIME

SETUP

$C = C_{max}^{\infty} e^{-kt}$, with t = 3 hr

After the last dose no more drug goes in, so the level falls from the steady-state peak by first-order elimination alone.

ALGEBRA

$e^{-kt} = e^{-(0.1733\ hr^{-1})(3\ hr)} = e^{-0.520} = 0.5945$

k = 0.1733 hr^{-1} and t = 3 hr, so the exponent is a pure number. The time is the 3 hours after the last dose, not the 8-hour interval.

ALGEBRA

$C = 53.3\ mg/L \times 0.5945 = 31.7\ mg/L$

Three hours is less than one half-life, so the level has fallen by less than half, and the answer must lie between 53.3 and 26.7 mg/L.

ROUND

$C = 31.7\ mg/L$

31.7 mg/L, three significant figures as in her other values. It is higher than 29.7 mg/L at the same time after the second dose, because more drug has accumulated by steady state.

> **DOES THE NUMBER MAKE SENSE?**
> Three hours is under one half-life of 4 hours, so more than half of the 53.3 mg/L peak remains; 53.3 × 0.5945 = 31.7 mg/L, below the peak and above half of it.

Slide "Plasma Drug Concentration at Steady State" Worked by the drill

Answer: 26.7 mg/L

WHAT THE STEM GIVES

t½	4 hours	gives $k = \frac{0.693}{4} = 0.1733\ hr^{-1}$; 4 hours is exactly one half-life
C_{max}^{∞}	53.3 mg/L	the peak the level decays from
t	4 hours after injection	one half-life, so $e^{-kt} = 0.5$

THE WORKING, ONE LINE AT A TIME

SETUP

$C = C_{max}{}^{\infty}e^{-kt}$, with t = 4 hr

After the last dose no more drug goes in, so the level falls from the steady-state peak, 53.3 mg/L, by first-order elimination.

ALGEBRA

4 hr = 1 half-life, so $C = \frac{53.3 \text{ mg/L}}{2}$

Four hours is exactly one half-life, so the level halves. With the exponential: $e^{-(0.693/4)(4)} = e^{-0.693} = 0.5$.

ALGEBRA

C = 26.7 mg/L

Half of 53.3 is 26.65. Using the exponential gives the same: $53.3 \times e^{-0.693} = 53.3 \times 0.5$, because kt = 0.693 exactly when t is one half-life.

ROUND

C = 26.7 mg/L

26.7 mg/L to three significant figures. After a second half-life, 8 hours after the last dose, it would be 13.3 mg/L, the steady-state trough.

> **DOES THE NUMBER MAKE SENSE?**
> Four hours after the dose is exactly one half-life of 4 hours, so the level is half the 53.3 mg/L peak: 53.3 × 0.5 = 26.7 mg/L, below the peak.

Example 4 Her key

Answer: 150 mg/hr

WHAT THE STEM GIVES

D_0	300-mg dose	the numerator
infusion time	over a period of 2 hours	the denominator

THE WORKING, ONE LINE AT A TIME

SETUP

$R = \frac{\text{dose}}{\text{infusion time}}$

A constant-rate infusion spreads the whole dose evenly over its duration, so the rate is the amount divided by the time. This R goes into the infusion equation.

ALGEBRA

$R = \frac{300 \text{ mg}}{2 \text{ hr}} = 150 \text{ mg/hr}$

Milligrams per hour is an amount per unit time, which is what a zero-order rate is. The equation needs the rate, not the dose.

ROUND

R = 150 mg/hr

Her value, 150 mg/hr. Every later part of this example uses this rate with k = 0.15 hr^{-1} and V_D = 15 L, so an error here carries into every later concentration.

DOES THE NUMBER MAKE SENSE?
300 mg delivered evenly over 2 hours is 150 mg each hour, so R = 150 mg/hr. The rate is above zero.

Example 4(a) Her key

Answer: 17.28 mg/L

WHAT THE STEM GIVES

D_0	300-mg dose	with the infusion time gives $R = \frac{300}{2} =$ 150 mg/hr
infusion time	over a period of 2 hours	$R = \frac{300}{2} = 150$ mg/hr; also t = 2 hr in e^{-kt}
k	0.15 hr^{-1}	in $Cl = kV_D$ and in the exponent $kt = 0.15 \times 2 = 0.30$
V_D	15 L	in $Cl = kV_D = 2.25$ L/hr

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_p = \frac{R}{V_D\, k}\,(1 - e^{-kt}), \text{ with } t = 2 \text{ hr}$$

During an infusion the concentration rises towards the plateau $\frac{R}{V_D\, k}$, with R the infusion rate, V_D the volume of distribution and k the elimination rate constant; t is the infusion time, 2 hr, not the dosing interval.

UNIT

$$R = \frac{300 \text{ mg}}{2 \text{ hr}} = 150 \text{ mg/hr};\ V_D\, k = (15 \text{ L})(0.15 \text{ hr}^{-1}) = 2.25 \text{ L/hr}$$

V_D times k is the clearance in L/hr, so R over it is mg/hr over L/hr, which leaves mg/L. When clearance is given directly, $\frac{R}{Cl}$ is the same quantity.

ALGEBRA

$$kt = (0.15 \text{ hr}^{-1})(2 \text{ hr}) = 0.30;\ 1 - e^{-0.30} = 1 - 0.7408 = 0.2592$$

Two hours is less than one half-life ($\frac{0.693}{0.15} = 4.6$ hr), so only about a quarter of the plateau is reached.

ALGEBRA

$$C_p = \frac{150 \text{ mg/hr}}{(15 \text{ L})(0.15 \text{ hr}^{-1})}\,(1 - e^{-(0.15)(2)}) = 66.67 \times 0.2592 = 17.28 \text{ mg/L}$$

The bracket $(1 - e^{-kt}) = 0.2592$ is the fraction of the plateau reached when the infusion stops, so C_p is the plateau $\frac{R}{V_D\, k} = 66.67$ mg/L times 0.2592: 17.28 mg/L, about 26 per cent of the plateau.

ROUND

C_p = 17.28 mg/L

Her value, 17.28 mg/L, which she carries as 17.3 into part (b). It is the starting value for the decline after the infusion.

> **DOES THE NUMBER MAKE SENSE?**
>
> The plateau R over Cl is $\frac{150}{2.25}$ = 66.67 mg/L; after 2 hours, with kt = 0.30, only $1 - e^{-0.30}$ = 0.2592 of it is reached, so the level is well below 66.67 mg/L: 17.28 mg/L.

Example 4(b) Her key

Answer: 13.34 mg/L

WHAT THE STEM GIVES

D_0	300-mg dose	*not needed: the end-of-infusion level is given*
infusion time	over a period of 2 hours	fixes when each infusion ends: 2 hr and 8 hr
second start	Six hours after the start of the first infusion	the second infusion runs 6 to 8 hr
k	0.15 hr^{-1}	the exponent of each decay term
V_D	15 L	*not needed: the end-of-infusion level is given*
Cend	17.28 mg/L	the starting level of each decay term
t	4 hours after the cessation of the second infusion	the second term decays 4 hr, the first 10 hr

THE WORKING, ONE LINE AT A TIME

SETUP

Time line: first infusion 0 to 2 hr; second infusion 6 to 8 hr; time asked for 8 + 4 = 12 hr

The second infusion starts 6 hours after the first started and runs 2 hours, so it ends at 8 hours; 4 hours later is 12 hours. Writing the times down first keeps the two exponents straight.

SETUP

$$C = 17.28\ e^{-k(12\ -\ 2)} + 17.28\ e^{-k(12\ -\ 8)}$$

Each infusion contributes its end-of-infusion concentration, declined for the time since its own end: 10 hours for the first, 4 hours for the second. The two add because the kinetics are first order.

ALGEBRA

First: $17.28 \times e^{-1.5} = 17.28 \times 0.2231 = 3.86$ mg/L

kt = 0.15 × 10 = 1.5, a pure number. Ten hours is more than two half-lives, so under a quarter of the first infusion is left.

ALGEBRA

Second: $17.28 \times e^{-0.6} = 17.28 \times 0.5488 = 9.48$ mg/L

kt = 0.15 × 4 = 0.6, a pure number. Four hours is less than one half-life (4.6 hr), so over half of the second infusion is left, which is why it dominates the sum.

ALGEBRA

$C = 3.86 + 9.48 = 13.34$ mg/L

The sum of the two contributions is the plasma concentration at 12 hours. Adding is allowed because elimination is first order, so each infusion declines independently of the other.

ROUND

$C = 13.34$ mg/L

Her stated value is 13.33 mg/L. Carrying 17.28 through gives 13.34, and starting from 17.3 gives 13.35; all three are within the tolerance.

DOES THE NUMBER MAKE SENSE?

The second infusion alone leaves 17.28 × 0.5488 = 9.48 mg/L after 4 hours, and the first adds what is left of 17.28 after 10 hours, so the sum lies between 9.48 and 2 × 17.28 mg/L.